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INTERIM FEDERAL SPECIFICATION

HEAD, SPRINKLER, (UNDERGROUND CONNECTED)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sprinkler heads for installation in fixed underground piping systems for watering lawns.

1.2 Classification. Sprinkler heads shall be furnished in the following types and classes, as specified (see 6.2).

1.2.1 Types.

Type I - Fixed Surface head
Type II - Pop-up head

1.2.2 Classes.

Class a - full circle pattern
Class b - half circle pattern
Class c - third circle pattern
Class d - quarter circle pattern
Class e - square pattern

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

FSC 4730

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3.1 Material. Sprinkler head bodies shall be cast of copper alloy or zinc base alloy. All internal parts shall be of wrought copper alloy.

3.2 Construction

3.2.1 Type I - Fixed surface head. Type I sprinkler heads shall be designed to be installed flush with the ground, and with a top flange, 2-1/8 inches minimum diameter, to keep the grass away from the spray nozzle. The sprinkler head shall be fully adjustable for flow and coverage and shall be suitably designed to produce a dense conical spray. The body shall be constructed with a 1/2 inch female thread for installation in a fixed underground pipe system.

3.2.2 Type II - Pop-up head. The type II sprinkler head shall be designed to be installed flush with the ground and equipped with a nozzle that rises a minimum of 1 inch above the body when the water is turned on. The rising member supporting the nozzle shall be identical on full, half, third or quarter pattern sprinklers so that nozzles will be interchangeable. The sprinkler head shall be designed to spray a dense, fine-spray water distribution pattern and shall be adjustable for coverage and flow. The nozzle shall be removable so sprinkler head does not have to be removed for flushing or cleaning. The body shall be constructed with a 1/2 inch female thread for installation in a fixed underground pipe system.

3.3 Performance. Types I and II, classes a through e sprinkler heads shall be capable of meeting the requirements of pattern types, water pressures, flow and area coverage, in accordance with table I.

Table I. Performance requirements

Type I - Fixed Surface sprinkler head

Class	Pattern type	Water pressure at sprinkler	Flow	Coverage
		p.s.i.	G.P.M	Feet (min)
a	Full circle	15	2.0-2.2	20 diam.
b	Half circle	15	1.5-1.7	10 radius
c	Third circle	15	1.0-1.2	10 radius
d	Quarter circle	15	0.8-1.0	10 radius
e	Square pattern	15	2.0-2.2	16 x 16

Type II - Pop-up sprinkler head

Class	Pattern type	Water pressure at sprinkler	Flow	Coverage
		p.s.i.	G.P.M.	Feet (min.)
a	Full circle	20	2.9-3.1	24 diam.
b	Half circle	20	2.0-2.2	12 radius
c	Third circle	20	1.0-1.8	12 radius
d	Quarter circle	20	0.9-1.5	12 radius
e	Square pattern	20	2.9-3.1	24 x 24

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3.4 Workmanship. Sprinkler heads shall be free of porosity, cold shuts, cracks or any other defects which may impair their serviceability. They shall be clean both inside and outside, and all fins and roughness shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type and class of sprinkler heads submitted for inspection at the same time.

4.2.2 Visual examination of end item. Sampling for visual examination shall be in accordance with level S-3 in MIL-STD-105. The Acceptable Quality Level (AQL's) for defects shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.2.3 Sampling for tests. Samples for tests shall be in accordance with level S-2 in MIL-STD-105. The AQL for tests shall be 6.5 expressed in defects per 100 units.

4.3 Examination

4.3.1 End item. Sprinkler heads selected in accordance with 4.2.2 shall be examined for defects listed in table II.

Table II. Classification of defects, end item

Defect	Major	Minor
Type and class not as specified.	X	
Porosity covering more than 10 percent of the surface area.	X	
Cracks or cold-shuts.	X	
Sharp edges, fins, projections.	X	
Surface rough.		X
Damage or defects not affecting function or serviceability.		X
Damage or defects affecting function or serviceability.	X	

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6 defects per hundred units.

Table III. Classification of preparation for delivery defects

Examine	
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship.	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.4 Tests

4.4.1 General. Sprinkler heads selected in accordance with 4.2.3 shall be tested to verify compliance with this specification.

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4.4.2 Water pressure. Water pressure for type I shall be regulated at 15 p.s.i. ± 2 p.s.i. and for type II at 20 p.s.i. ± 2 p.s.i.

4.4.3 Flow. Rate of flow shall be regulated within ± 0.1 g.p.m. (see table I).

4.4.4 Coverage and spray pattern. Tests shall be conducted with the sprinkler heads mounted on a suitable fixture, at ground level, on a surface such as concrete or blacktop, so that water coverage may be easily measured and the spray pattern observed.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The sprinklers shall be cleaned by process C-3 and dried by procedure D-1 in accordance with MIL-P-116.

5.1.1.2 Unit packaging. Unless otherwise specified, 12 sprinklers shall be packaged by method IC-2 in accordance with MIL-P-116.

5.1.2 Level B. Twelve sprinklers shall be packaged in a close-fitting box conforming to PPP-B-566, style VIII, type A, class a; PPP-B-665, style A or PPP-B-676, type I. Cushioning shall be provided and the box closed in accordance with the appendix to the applicable box specification.

5.1.3 Level C. The sprinklers shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Sprinklers of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, type overseas; PPP-B-621, class 2; PPP-B-636, class weather resistant, or PPP-B-640, class 2, grade A. The fiberboard boxes shall be closed in accordance with the appendix to the box specification. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed the weight limitations of the box specification. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Eight packages of sprinklers of like description packaged as specified in 5.1.2 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The sprinklers shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.2.4 Standard pack (civil agency procurements). The standard pack for civil agency procurements shall be as specified in paragraphs 5.1.2 and 5.2.2 (see 6.3).

5.3 Marking

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Sprinkler heads are intended to be permanently connected, at surface level, to an underground piping system, used for irrigating lawns.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of sprinkler head required (see 1.2.1 and 1.2.2).
(c) Size of lot, if different from 4.2.1.
(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 Standard pack for civil agencies. The standard pack requirements in 5.2.4 are intended for use in procurement of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot of intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

MILITARY INTERESTS:

Army - MO
Navy - Ships
Air Force - 69

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